

Personal Information

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Profile

I started my career as a system engineer back in 2008, and over the years built up deep fullstack engineering experience across React, Node.js, Python, and cloud platforms. That hands-on breadth is what I now bring to AI engineering as the founder of Elands AI, where we design and deliver intelligent systems for startups and small companies.

As a Dutch citizen and fluent speaker of English, Dutch and Serbian, I have worked extensively with Dutch 🇳🇱, American 🇺🇸 and British 🇬🇧 customers.

Building with generative AI is no longer just a learning interest: it is my day job. Through Elands AI, I design and ship LLM-based applications, evaluation infrastructure, and real-time AI services for clients. You can browse some of my experiments on [GitHub](#), including [agentic usage with MCP servers](#).

I spend my free time with family, learning Dutch, exploring emerging technologies and taking care of my health via balanced diet and strength training.

And as a Dutchman, I ride kids to school in our Urban Arrow bakfiets (cargobike) - so fun and "gezellig" 🎉 (joyful in Dutch)

Professional experience

From 06/2025 - now **Lead AI Engineer at Delphyr B.V.**

- At Delphyr AI I founded the engineering effort behind a next-gen AI platform built specifically for the medical field, and led it through its first year. We are bringing AI into real clinical workflows, from hospitals to GP practices, without compromising on usability, privacy, or compliance.
- I now focus on the AI layer: prompt design, safety and grounding evaluation, and the observability that keeps model behavior measurable in production. I work mainly in Python and FastAPI, across the Kubernetes and TypeScript surfaces around it.
- Architecting and building the adversarial guardrail evaluation suite our safety layer is graded against: jailbreaks, prompt extraction, medical misinformation, balanced against safe controls
- Owning prompt design and grounding for ambient clinical documentation, from per-specialism templates to transcript-level verification of every generated statement
- Building production-grade LLM systems: from evaluation harness and release gating on measured regressions to LangWatch observability and PHI-safe tracing

From 09/2024 - 06/2025 **Expert Software Engineer at Totally Gifts**

- Development of a greenfield project with a new tech stack, from scratch
- NextJS, TailwindCSS, Clerk, Sentry, NestJS, PostgreSQL, Docker, AWS have been used to build the solution
- Event-driven architecture was used to capture gift card events and process them in real-time

- I contributed across the stack, collaborating closely with a cross-functional team of three engineers, a designer, and a product owner
- Management of App Store and Play store releases

From 11/2023 - 09-2024 **Lead Fullstack Software Engineer at Grndhouse**

- Team lead managing a team of 5 backend and frontend developers on grndhouse.com, a platform for on-demand strength training.
- Led the development of the initial MVP after a successful funding round
- Took regular meetings with stakeholders to streamline delivery of video streaming, payment processing, and user management features
- Architected and implemented features using React Native, Node.js, and AWS
- Mentored junior developers, resolved blockers, and ensured timely project delivery
- Used a wide array of modern tools and services:
 - o Sentry for event monitoring
 - o RevenueCat for payments
 - o Mixpanel for analytics
 - o Expo to manage app lifecycle and deployments with or without App Store/Play Store releases

From 12/2021 - 11/2023 **Fullstack Software Engineer at Fertifa**

- Progressed from individual contributor to team lead, managing a team of 5 backend and frontend developers on fertifa.com, Europe's most comprehensive reproductive healthcare provider.
- Defined and developed features in collaboration with the product owner and stakeholders
- Engineered solutions in React Native, Node.js, and AWS
- Mentored junior developers and led sprints to ensure timely delivery
- Tools and services used:
 - o Strapi for content management
 - o Sentry for monitoring
 - o Google Analytics, Meta Pixel, Drip, Amplitude for analytics
 - o Manual app lifecycle management and deployments

06/2018 – 12/2021 **Senior Software Engineer at KPN Technium B.V.**

- I was senior engineer in team that was developing ETL automation framework and platform in Azure, based on Hadoop & Kafka ecosystem, automated by Terraform
- The rendering framework was built with Python, Flask and SQLAlchemy
- Led story mapping for migration of on-premise data platform services to Azure
- Development of distributed event sourcing application for “real-time” analysis of KPN customer journey where multiple touch points are combined, analysed and visualised
- Scala stack with Akka/Alpakka is used to build microservices and Flink applications
- Utilizing DevOps methodology with fully automated CI/CD, monitoring and alerting

02/2017 – 05/2018 **Software Engineer, Cognitive Implementation at IPsoft B.V.**

- As technical lead of a 3-member agile team I was responsible for development of human-machine interaction on artificial intelligence framework “Amelia”.
- I was mostly busy with hands-on development, code reviewing, mentoring and Scrum rituals.
- Development was mostly in Groovy/Python and in Java
- The technical stack included Kafka, Camel, Mule, Elasticsearch, Spring, Grails, Docker
- Data analysis with Pandas, Jupyter and visualisation with d3.js, and making new features (or refactoring) decisions from the insights

06/2012 – 01/2017 **Senior System Engineer at Liberty Global B.V.**

- As member of Liberty Global OTT streaming solution architecture & engineering team, I was busy with architecture of various system components, developing AWS (micro)services infrastructure, CDN content delivery, Adobe AEM and their integration
- Microservices development in Python
- Through work with high availability services I acquired deep knowledge of web and backend application design, streaming technologies, HA scalable infrastructure, AWS cloud and full stack network protocols
- Data analysis with Jupyter, Wireshark, Conviva, Omniture
- Performing code reviews and overall system troubleshooting

Technical knowledge and other skills

AI and LLM systems

- LLM APIs (OpenAI-compatible: Azure ML, Nebius; Google Gemini), MCP servers and agentic tool use, Pydantic AI and pydantic-evals, prompt design and grounding, LLM-as-judge evaluation, guardrails and adversarial testing, LangWatch, PHI de-identification (Deduce)

Languages

- Python, TypeScript, Scala, Java

Frameworks and libraries

- FastAPI, Pydantic, SQLAlchemy/Alembic, PostgreSQL, Node.js/NestJS, Kafka, Spark/Databricks, Flink, pandas
- React, React Native, Next.js, TailwindCSS, Playwright, Jest
- Kubernetes, Docker, ArgoCD, AWS, Azure, Terraform, GitHub Actions, OpenTelemetry, Prometheus/Grafana, Sentry

Languages

- Fluent in English, Dutch (NT2) and Serbian.
- Intermediate level of German, Russian.

Education

- 09/2009 – 02/2011 **Masters Course: System Engineering and Radio Communications** – Faculty of Electrical Engineering, University of Belgrade. Specialization: Telecommunications and software engineering
- 10/2002 – 08/2007 **BSc of Electrical Engineering** - Faculty of Electrical Engineering, University of Belgrade. Specialization: Telecommunications and computer science

Certificates

- Acclaim badges: <https://www.credly.com/users/dejan-petkovic/badges>
- Coursera certificates: <https://coursera.org/share/32a42acb44082359b7c10bfea2f765ee>

Projects

- BrAln: <https://github.com/dexpetkovic/brAln>
 - o WhatsApp-based AI assistant built with NestJS and Google Gemini. Handles WhatsApp webhook events and generates contextual replies, with an MCP server that lets the model store, update and retrieve memories.
- Biller: <https://biller.elands.studio>
 - o Professional invoicing for freelancers. Create, send, and track invoices in seconds. One-time purchase, no subscriptions.
- Bouwen: <https://bouwen.elands.studio>

- o Renovation project communication for the Dutch market. Print a QR code for your window and let neighbours follow your renovation in real time.
- 2e-woning.nl: <http://2e-woning.nl>
 - o Box 3 tax calculator for a second home and other investments. Built for the Dutch market.